Polynomial identities

1 Follow the steps below to prove that $\left(a^2+b^2\right)\left(c^2+d^2\right)=(ac-bd)^2+(bc+ad)^2$.

a Distribute the left-hand side of the equation.

b Distribute the right-hand side of the equation.

2 Prove $(m-n)^2=m^2-2mn+n^2$.

3 Use the identity $(u+v)^2=u^2+2uv+v^2$ with $v<10$ to find the value of 1002^2.

4 Prove $x^3-y^3=(x-y)\left(x^2+xy+y^2\right)$.

5 Use the identity $m^4-n^4=(m-n)(m+n)\left(m^2+n^2\right)$ to find the value of 8^4-2^4.

6 Use the identity $u^6+v^6=\left(u^3-2uv^2\right)^2+\left(v^3-2u^2v\right)^2$ to find the value of 5^6+2^6.

7 Consider the difference of two squares expression u^2-v^2.

a Express the difference of two squares as a product.

b Use the identity $u^2-v^2=(u+v)(u-v)$ to find the value of 77^2-23^2.

c Use the identity $u^2-v^2=(u+v)(u-v)$ to find the value of 24×16.

8 Prove $x^5+y^5=(x+y)\left(x^4-x^3y+x^2y^2-xy^3+y^4\right)$.

9 Prove $m^6+n^6=\left(m^3-2mn^2\right)^2+\left(n^3-2m^2n\right)^2$.

10 Use the identity $\left(a^2+b^2\right)\left(x^2+y^2\right)=(ax-by)^2+(bx+ay)^2$ to find the value of $\left(27^2+11^2\right)\left(42^2+94^2\right)$.

Applications

11 Kathleen noticed that if she multiplies the square number 289 by another square number 64, the result is a square number.

a What is the square root of the resulting product Kathleen found?

b Let x^2 be a square number, and y^2 be different square number. Show that the product of two square numbers is always a square number.

12 Consider the sum of consecutive odd numbers.

a Evaluate the sum $1+3+5+7$.

b Evaluate the sum of the first 7 odd numbers.

c Form a simplified expression for the sum of the first t odd numbers: $1+3+5+...+(2t-1)$.

d Using this result, find the sum of the first 70 odd numbers.

e Can 3972 be the sum of consecutive odd numbers (starting from 1)?

13 Let mn be any two-digit number, where m is the tens digit and n is the unit digit.

a Express the value of the two-digit number in terms of m and n.

b The digits are switched to form a new two-digit number. Express the value of the new two digit number in terms of m and n.

c Show that the difference between the original two-digit number and the new two-digit number is always a multiple of 9.

14 Consider the product of three consecutive integers.

a 4080 is the product of three consecutive integers. Find the three integers using trial and error.

b The product of any three consecutive integers in which x is the first integer is given by $x\left(x+1\right)\left(x+2\right)$. Which of the following numbers will the product always be divisible by?

A 5 **B** 8 **C** 3 **D** 6

15 Consider the following.

a Expand and simplify $\left(m^2+n^2\right)^2$.

b Prove that $\left(m^2-n^2\right)^2+(2mn)^2=\left(m^2+n^2\right)^2$

c What special relationship is this related to?

d Using the identity proven, find the other two integer length sides of a right-angled triangle in which one shorter side measures 30 units.

16 The identity $\left(x^2+y^2\right)^2=\left(x^2-y^2\right)^2+(2xy)^2$ can be used to generate Pythagorean triples. Find the side lengths of a right-angled triangle using $x=8$ and $y=6$.

17 Consider the expression $(x-a)\left(x^2+ax+a^2\right)$.

a Distribute and simplify the expression.

b Show that $999\,973$ is not prime by expressing it as the product of two factors.

18 Consider the expression $\left(2x^2+x+2\right)\left(x^2+2x+2\right)$.

a Distribute and simplify the expression.

b Now, evaluate 212×122.

c If the identity from part (a) is to be used to evaluate the products of three digit numbers, what value does x represent?

19 Consider the following pattern:

$$\begin{aligned}4^2&=3^2+3+4\\5^2&=4^2+4+5\\6^2&=5^2+5+6\end{aligned}$$

We wish to show that this pattern will continue for any square number. Let x^2 be the square number.

a According to the pattern, how can x^2 be rewritten?

b Prove that $x^2=(x-1)^2+(x-1)+x$.

c Using the identity proven in the previous part, evaluate 101^2.

20 Consider the expression $2 \times 3 \times 4 \times 5 + 1$.

a Evaluate $2 \times 3 \times 4 \times 5 + 1$.

b Find the missing value:

$$2 \times 3 \times 4 \times 5 + 1 = (\square)^2$$

c We want to show that 1 more than the product of four consecutive integers is always a perfect square. Let x be the first of any four consecutive integers. Complete the following work to show that $x\,(x+1)\,(x+2)\,(x+3)+1$ can be expressed as the square of a value:

$$\begin{aligned} x\,(x+1)\,(x+2)\,(x+3)+1 &= \left(x^2+x\right)\left(x^2+\square x+\square\right)+1 \\ &= x^4+\square x^3+\square x^2+\square x+\square \\ &= \left(x^2+\square x+1\right)\left(x^2+\square x+1\right) \\ &= \left(x^2+\square x+1\right)^{\square} \end{aligned}$$